

KE4CON · FieldCommand IMS

McHenry County Emergency Services Volunteers · K9ESV

FIELDCOMMAND

Incident Management System v1.0

BETA TEST SCENARIOS

Six Step-by-Step Scenarios · All 49 Tools

Scenario 1: Setup · 2: Personnel · 3: ICS · 4: Amateur Radio · 5: FEMA · 6: Multi-Op

July 24, 2026

FOR BETA TESTING USE ONLY — NOT FOR OPERATIONAL USE

This scenario tests the complete first-run experience: connecting to FieldCommand IMS for the first time, configuring the organization identity, verifying the dashboard in all three modes, creating a new incident from scratch and from a pre-planned template, and testing the archive, restore, and scenario reset functions. Complete this scenario first before running any other.

■ Use a clean FieldCommand system with no prior data for this scenario. If the system has been used previously, run Beta Reset first (incident_mgmt.html → Beta/Scenario Reset) before starting.

PRE-TEST CHECKLIST — Verify the following before starting:

- FieldCommand Pi 5 is powered on and has been running for at least 60 seconds
- Your test device (phone, tablet, or laptop) is connected to EMCOMM-NET Wi-Fi
- Browser is open to <http://192.168.50.1>
- No active incidents exist (or Beta Reset has been performed)
- Test roster CSV is prepared with at least 5 members (see Appendix A)
- Note the current time — you will need it for incident timestamps

TEST 1 — Organization Setup (setup.html)

The Organization Setup page configures the station identity that appears on all ICS forms, net logs, and printed documents. Test that all fields save correctly and persist after a browser refresh.

Step 1. Navigate to the Organization Setup page. Navigate to: http://192.168.50.1/setup.html	→ <i>Expected:</i> Page loads with empty or default fields.
Step 2. Enter your organization name: <i>Beta Test Emergency Services</i>	→ <i>Expected:</i> Field accepts text input.
Step 3. Enter short name: <i>BTES</i>	→ <i>Expected:</i> Field accepts 4-character abbreviation.
Step 4. Enter primary callsign: <i>W8TEST</i>	→ <i>Expected:</i> Field accepts callsign format.
Step 5. Enter agency name: <i>Beta County Emergency Management</i>	→ <i>Expected:</i> Field accepts text.
Step 6. Enter city: <i>Testville</i> and state: <i>IL</i>	→ <i>Expected:</i> Location fields accept input.
Step 7. Enter grid square: <i>EN52ab</i>	→ <i>Expected:</i> Grid square field accepts 6-character Maidenhead format.
Step 8. Set time zone to your local time zone.	→ <i>Expected:</i> Dropdown shows all US time zones.
Step 9. Click Save Configuration .	→ <i>Expected:</i> Success message appears. No error.
Step 10. Press F5 to refresh the browser. Return to setup.html. Navigate to: http://192.168.50.1/setup.html	→ <i>Expected:</i> All fields still show the values you entered — data persisted to database.
Step 11. Navigate to the main dashboard. Navigate to: http://192.168.50.1/index.html	→ <i>Expected:</i> Dashboard header shows your organization name.
■ Record any fields that do NOT persist after refresh.	

TEST 2 — Dashboard Navigation & WAN Status (index.html)

Step 12. On the main dashboard, verify the WAN status bar is visible at the top. Navigate to: http://192.168.50.1/index.html	→ <i>Expected:</i> WAN status bar shows — either green (Cellular), blue (Satellite), or red (Offline). Color matches actual internet availability.
Step 13. Click the All-Hazards ICS mode tab.	→ <i>Expected:</i> Tool card grid changes to show ICS platform tools: Incident Management, T-Card Board, ICS Forms, FEMA costs, etc.
Step 14. Click the Amateur Radio EMCOMM mode tab.	→ <i>Expected:</i> Grid shows Net Control Logger, FCC Callsign Lookup, APRS Map, Winlink, JS8Call, etc.
Step 15. Click the Public Safety mode tab.	→ <i>Expected:</i> Grid shows Starcom Net Logger, Resource Map, WAN Status, Weather Radar, Hospitals.

Step 16. Click every tool card visible in any mode and confirm each page loads. Note any cards that produce a 404 or error.	→ <i>Expected:</i> All cards navigate to the correct page. No 404 errors.
Step 17. Wait 30 seconds and observe the WAN status bar.	→ <i>Expected:</i> Status bar refreshes automatically without a manual page reload.

TEST 3 — Preflight Deployment Checklist (preflight.html)

The Preflight page auto-checks six data readiness items on page load: organization setup, FCC database (count + age), roster member count, repeater database (count + coordinates), channel library, and active incident. Verify both the auto-checks and the manual sections.

Step 18. Navigate to the Preflight Check page. Navigate to: <code>http://192.168.50.1/preflight.html</code>	→ <i>Expected:</i> Page loads. Data Readiness section auto-runs 6 checks: org setup, FCC DB, roster count, repeater DB (with coordinate count), channel library, active incident.
Step 19. Review the Data Readiness auto-check results.	→ <i>Expected:</i> Each item shows ✓ (green) or ✗ (red) with count/detail. Items with issues link directly to the fix page.
Step 20. Work through each manual checklist section (Power, Comms, Computing, Personnel, Logistics, Safety, Coordination). Check every applicable item.	→ <i>Expected:</i> Each checkbox toggles and remains checked.
Step 21. Add a note in the Notes field: <i>Beta test run — all hardware nominal</i>	→ <i>Expected:</i> Note field accepts free text.
Step 22. Click Save Preflight or submit the form.	→ <i>Expected:</i> Success confirmation. Data saved.
Step 23. Refresh the page.	→ <i>Expected:</i> Checked items and notes persist. Auto-checks re-run on reload.
Step 24. Click Print .	→ <i>Expected:</i> Print dialog opens with a clean single-page preflight layout.

TEST 4 — Pre-Planned Event Templates (event_templates.html)

Step 25. Navigate to Event Templates. Navigate to: <code>http://192.168.50.1/event_templates.html</code>	→ <i>Expected:</i> Six built-in templates visible: Emergency Shelter, SAR, Severe Weather, Mass Gathering, HazMat, Exercise.
Step 26. Click on the Exercise / Training Scenario template card.	→ <i>Expected:</i> Configuration panel opens with incident name field and template details.
Step 27. Enter incident name: <i>Beta Test Exercise Alpha</i> . Click Activate Template .	→ <i>Expected:</i> New incident created. Browser navigates to incident dashboard. Incident marked with ■■ scenario badge.
Step 28. Verify the ICS-202 objectives pre-filled from the template.	→ <i>Expected:</i> Training objectives appear in the incident objectives list.
Step 29. Return to Event Templates. Click Manage Templates tab. Navigate to: <code>http://192.168.50.1/event_templates.html</code>	→ <i>Expected:</i> All six templates listed with Edit buttons.
Step 30. Click Edit on Emergency Shelter Operations.	→ <i>Expected:</i> Edit modal opens with four tabs: Objectives, Resources, Channels, Organization.
Step 31. Add a new objective: <i>Establish shelter registration within 30 minutes</i> . Save.	→ <i>Expected:</i> Objective appears in the list. Save succeeds.
Step 32. Click Export JSON on any template.	→ <i>Expected:</i> JSON file downloads to your device.
Step 33. Click Import JSON and import the file you just exported.	→ <i>Expected:</i> Template imports successfully. Appears in list.

TEST 5 — Manual Incident Creation (incident.html)

Step 34. Navigate to Incident Management. Navigate to: <code>http://192.168.50.1/incident.html</code>	→ <i>Expected:</i> Any existing incidents are listed.
Step 35. Click + New Incident .	→ <i>Expected:</i> New incident form opens.
Step 36. Enter: Name = <i>Scenario 1 Test Incident</i> , Type = <i>All-Hazards</i> , IC = <i>Beta Tester One</i>	→ <i>Expected:</i> Fields accept input.
Step 37. Click Create Incident .	→ <i>Expected:</i> Incident opens immediately. Incident name appears in dashboard header.
Step 38. Open a new operational period. Click + New Period .	→ <i>Expected:</i> Period 2 created. Period selector updates.
Step 39. Switch back to Period 1. Verify Period 1 data still shows.	→ <i>Expected:</i> Data is period-specific and switching periods changes the displayed data.
Step 40. Create a second incident with type <i>Search and Rescue</i> .	→ <i>Expected:</i> Second incident created and listed alongside the first.
Step 41. Set the first incident as Active by clicking it and selecting Set Active .	→ <i>Expected:</i> Active incident badge updates in dashboard header.

TEST 6 — Incident Archive, Restore & Hard Delete (incident_mgmt.html)

■ A USB drive labelled FIELDCOMMAND must be connected to the Pi for archive/restore tests. If no USB is available, skip Steps 42–46 and note as skipped.

Step 42. Navigate to Incident Management. Navigate to: <code>http://192.168.50.1/incident_mgmt.html</code>	→ <i>Expected:</i> All created incidents are listed.
Step 43. Select the SAR incident. Click Archive to USB .	→ <i>Expected:</i> Success message confirms archive written to <code>/media/fieldcommand/backup/incidents/</code> on USB.
Step 44. Verify the USB drive — confirm the JSON archive file exists at <code>/media/fieldcommand/backup/incidents/</code> on the Pi.	→ <i>Expected:</i> JSON file present with incident name in filename.
Step 45. Click Hard Delete from Pi on the archived SAR incident. Confirm the deletion prompt.	→ <i>Expected:</i> Incident removed from the Pi database. No longer listed.
Step 46. Click Restore from USB . Select the JSON file you archived.	→ <i>Expected:</i> Incident restored to fully active status. All data intact.
Step 47. Navigate to Incident Management. Click Beta/Scenario Reset . Navigate to: <code>http://192.168.50.1/incident_mgmt.html</code>	→ <i>Expected:</i> Confirmation prompt appears warning this will wipe all incident data.
Step 48. Confirm the Beta Reset.	→ <i>Expected:</i> All incidents, forms, costs, check-ins, and T-cards wiped. Roster, hospitals, channels, and repeaters preserved.
Step 49. Navigate to roster.html to verify roster is intact after reset. Navigate to: <code>http://192.168.50.1/roster.html</code>	→ <i>Expected:</i> Roster still shows any members previously entered.

TEST 7 — WAN Source Configuration (wan_settings.html)

FieldCommand supports any WAN source your group chooses — cellular modem, satellite dish, site Ethernet, USB tether, or venue Wi-Fi. The WAN Settings page lets you configure which source is preferred, which is fallback, and how each is detected.

Step 50. Navigate to WAN Settings. Navigate to: <code>http://192.168.50.1/wan_settings.html</code>	→ <i>Expected:</i> Current WAN sources listed with their roles (Preferred/Fallback).
Step 51. Note the current Preferred source. Click ■ Swap Roles .	→ <i>Expected:</i> Preferred and Fallback roles swap immediately.
Step 52. Swap roles back.	→ <i>Expected:</i> Original configuration restored.

Step 53. Change the detection method on one source to <i>ping</i> . Enter a gateway IP (use 8.8.8.8 for testing).	→ <i>Expected</i> : Detection method updates. Field for gateway IP appears.
Step 54. Save the change. Wait 60 seconds.	→ <i>Expected</i> : WAN status on dashboard updates to reflect new detection.
Step 55. Disable one source using the Enabled toggle.	→ <i>Expected</i> : Source shows as disabled. WAN monitor no longer checks it.
Step 56. Re-enable the source.	→ <i>Expected</i> : Source re-enabled. WAN monitor resumes.

SCENARIO 1 — PASS / FAIL SUMMARY

TEST	RESULT	NOTES / DEFECTS
Test 1 — Organization Setup	■ Pass ■ Fail ■ Skip	
Test 2 — Dashboard Navigation	■ Pass ■ Fail ■ Skip	
Test 3 — Preflight Checklist	■ Pass ■ Fail ■ Skip	
Test 4 — Event Templates	■ Pass ■ Fail ■ Skip	
Test 5 — Incident Creation	■ Pass ■ Fail ■ Skip	
Test 6 — Archive / Restore	■ Pass ■ Fail ■ Skip	
Test 7 — WAN Settings	■ Pass ■ Fail ■ Skip	

Overall Scenario Result:

■ PASS — All tests passed or only minor cosmetic issues noted ■ CONDITIONAL — Passes with workaround (document defects above) ■ FAIL — Critical defect found (document and halt further testing)

Tester (name / callsign): _____

Date: _____

Time: _____

✓ Proceed to Scenario 2 only after this scenario passes.

BETA TEST SCENARIOS

SCENARIO	DURATION	FEATURES TESTED	PAGES EXERCISED
2 of 6	60–90 min	Roster · QR codes · Manual check-in · Scan check-in · PAR · Checkout · Cross-reference · Dead Man's Switch	roster.html · checkin.html · scan_checkin.html · accountability.html · deadmans.html · position_checklists.html

FOR BETA TESTING USE ONLY — NOT FOR OPERATIONAL USE

This scenario stress-tests every aspect of personnel management: building and importing the member roster, generating QR check-in codes, checking personnel in and out using both the manual form and the QR/barcode scanner, conducting a Personnel Accountability Report (PAR), and testing the Dead Man's Switch. Minimum 5 test personnel required.

■ **PREREQUISITE:** An active incident must exist from Scenario 1 (or create a new one). You need at least 5 people available as test subjects — they can be co-located or you can simulate them by creating roster entries and checking them in yourself.

TEST 1 — Member Roster (roster.html)

Step 1. Navigate to the Member Roster. Navigate to: <code>http://192.168.50.1/roster.html</code>	→ <i>Expected:</i> Roster page loads. Empty if first run.
Step 2. Click + Add Member . Enter: Member ID = ESV-001, First Name = Alpha, Last Name = Tester, Callsign = KD9TEST, Role = Net Control.	→ <i>Expected:</i> Member saves successfully. Appears in roster list.
Step 3. Add a second member WITHOUT a callsign (non-licensed): ID = ESV-002, Name = Bravo Tester, Role = Logistics.	→ <i>Expected:</i> Non-licensed member saves successfully without callsign.
Step 4. Add a third member with both callsign AND radio ID: ID = ESV-003, Callsign = KD9BETA, Radio ID = 412.	→ <i>Expected:</i> Member with both identifiers saves correctly.
Step 5. Click the QR button on ESV-001's card.	→ <i>Expected:</i> QR modal opens. QR code renders (online) or ID displayed (offline).
Step 6. Click ■ Print on the QR modal.	→ <i>Expected:</i> Print preview opens with QR code, member name, and check-in URL.
Step 7. Click ■ Save PNG .	→ <i>Expected:</i> PNG file downloads to your device.
Step 8. Click ■ Import CSV . Use CSV with columns <code>member_id</code> , <code>callsign</code> , <code>first_name</code> , <code>last_name</code> , <code>role</code> and 2 rows: ESV-004/KD9FOUR/Delta/Tester/Operator and ESV-005/Echo/Tester/Logistics.	→ <i>Expected:</i> Both members import successfully. Roster now shows 5 members.
Step 9. Click ■ Export CSV .	→ <i>Expected:</i> CSV file downloads containing all 5 roster members.
Step 10. Search for "Tester" using the roster search field.	→ <i>Expected:</i> All 5 members matching "Tester" appear. Non-matches hidden.
Step 11. Edit ESV-001. Check certifications: ICS-100, ICS-200, ICS-700.	→ <i>Expected:</i> Certification checkboxes save. Badges appear on roster card.

TEST 2 — Manual ICS-211 Check-In (checkin.html)

Step 12. Navigate to Incident Check-In. Navigate to: <code>http://192.168.50.1/checkin.html</code>	→ <i>Expected:</i> Check-in form loads with current incident and period.
Step 13. Enter callsign <i>KD9TEST</i> in the Callsign/ID field.	→ <i>Expected:</i> Name auto-fills from roster: "Alpha Tester".
Step 14. Select ICS Position: Net Control. Click Check In .	→ <i>Expected:</i> ESV-001 appears in the active check-in list below the form.
Step 15. Check in ESV-002 using radio ID 412.	→ <i>Expected:</i> Bravo Tester appears in check-in list.
Step 16. Check in ESV-003 using member ID ESV-003.	→ <i>Expected:</i> ESV-003 appears in check-in list.
Step 17. Check in a walk-in visitor NOT in the roster: Name = Victor Visitor, Agency = Red Cross, Position = Volunteer.	→ <i>Expected:</i> Visitor check-in succeeds without a roster match.
Step 18. In the active check-in list, click Check Out on Victor Visitor.	→ <i>Expected:</i> Victor Visitor moves to Checked Out section. Timestamp recorded.

Step 19. Refresh the page.	→ <i>Expected:</i> All check-in records persist. Checked-out entries remain in list.
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TEST 3 — QR/Barcode Scan Check-In (scan_checkin.html)	
■ This test requires a device with a camera (smartphone or tablet). Use the QR code PNG saved in Test 1, or display it on another screen. If no camera device is available, skip Steps 20–24 and test the manual fallback only.	
Step 20. Navigate to Scan Check-In on a camera-equipped device. Navigate to: <code>http://192.168.50.1/scan_checkin.html</code>	→ <i>Expected:</i> Page loads with camera area and manual entry fallback field.
Step 21. Click ■ Start Camera .	→ <i>Expected:</i> Camera activates. Viewfinder shows. Scan frame appears.
Step 22. Hold the QR code for ESV-004 in front of the camera.	→ <i>Expected:</i> Code detected automatically using BarcodeDetector API. Name "Delta Tester" auto-fills with green borders.
Step 23. Click ✓ Check In .	→ <i>Expected:</i> Full-screen success confirmation appears with name and time. Device vibrates briefly (if supported).
Step 24. Click Scan Next Person . Test manual entry fallback: type ESV-005 in the manual entry field and press Enter.	→ <i>Expected:</i> "Echo Tester" auto-fills from roster. Check-in completes.
Step 25. Scan a code NOT in the roster (use any random barcode — cereal box, ID badge, etc.).	→ <i>Expected:</i> Not-found banner appears. Manual form shown with scanned code pre-filled. Name field blank for manual entry.
Step 26. Verify session history panel shows the last 10 check-ins.	→ <i>Expected:</i> Recent check-ins listed at bottom of page.

TEST 4 — Personnel Accountability Report / PAR (accountability.html)	
Step 27. Navigate to Personnel Accountability. Navigate to: <code>http://192.168.50.1/accountability.html</code>	→ <i>Expected:</i> Page loads. Select the active incident.
Step 28. Review the summary tiles at the top.	→ <i>Expected:</i> Total Personnel, Checked In, Checked Out counts match actual check-in data.
Step 29. Click the ICS-211 CHECK-IN LIST tab.	→ <i>Expected:</i> All checked-in personnel displayed. Unconfirmed in default sort order.
Step 30. Click ■ Conduct PAR .	→ <i>Expected:</i> PAR timestamp recorded in header badge. Unconfirmed personnel sorted to top.
Step 31. Click ✓ PAR on ESV-001.	→ <i>Expected:</i> ESV-001 moves to PAR confirmed state. Green border appears. Count updates.
Step 32. Click ■ Location on ESV-002. Enter: Division Alpha, east staging.	→ <i>Expected:</i> Location saved. Shows under ESV-002's entry.
Step 33. Click ■ UNACCOUNTED tab.	→ <i>Expected:</i> Personnel not yet PAR-confirmed are listed with red background.
Step 34. Click T-CARD PERSONNEL tab.	→ <i>Expected:</i> Grouped by resource. Each resource shows PAR count (e.g. 0/3 PAR).
Step 35. Click CROSS-REFERENCE tab.	→ <i>Expected:</i> Shows personnel checked in but not on T-cards, and T-card personnel without ICS-211 check-in.
Step 36. Click ■ Reset PAR and confirm.	→ <i>Expected:</i> All PAR confirmations cleared. PAR badge resets.
Step 37. Click ■ Print .	→ <i>Expected:</i> Print dialog opens. Print buttons and controls hidden in print view.
Step 38. Enable Auto 60s checkbox. Wait 60 seconds.	→ <i>Expected:</i> Data refreshes automatically without manual reload.

TEST 5 — Dead Man's Switch (deadmans.html)

Step 39. Navigate to the Dead Man's Switch page. Navigate to: <code>http://192.168.50.1/deadmans.html</code>	→ <i>Expected:</i> Page loads with net selector and interval configuration.
Step 40. Set check-in interval to 1 minute for testing.	→ <i>Expected:</i> Interval field accepts numeric input.
Step 41. Click Start Monitoring .	→ <i>Expected:</i> Countdown timer begins. Shows time remaining before alert.
Step 42. Wait 60 seconds without logging a check-in.	→ <i>Expected:</i> Audible alert fires. Page flashes red. Alert message displayed.
Step 43. Go to Net Control Logger and log a check-in. Return to Dead Man's Switch.	→ <i>Expected:</i> Timer resets automatically after check-in is logged.
Step 44. Click Stop Monitoring .	→ <i>Expected:</i> Timer stops. Page returns to idle state.

TEST 6 — ICS Position Checklists (position_checklists.html)

Step 45. Navigate to ICS Position Checklists. Navigate to: <code>http://192.168.50.1/position_checklists.html</code>	→ <i>Expected:</i> Checklist page loads with position selector.
Step 46. Select Incident Commander (IC) position.	→ <i>Expected:</i> IC activation checklist displays.
Step 47. Check off the first 5 items on the IC checklist.	→ <i>Expected:</i> Items check and remain checked.
Step 48. Select a different position (e.g. Safety Officer).	→ <i>Expected:</i> SOFR checklist loads. IC checklist state saved separately.
Step 49. Click Print .	→ <i>Expected:</i> Print-ready checklist layout opens.

SCENARIO 2 — PASS / FAIL SUMMARY

TEST	RESULT	NOTES / DEFECTS
Test 1 — Member Roster	■ Pass ■ Fail ■ Skip	
Test 2 — Manual ICS-211 Check-In	■ Pass ■ Fail ■ Skip	
Test 3 — QR/Barcode Scan Check-In	■ Pass ■ Fail ■ Skip	
Test 4 — Personnel Accountability	■ Pass ■ Fail ■ Skip	
Test 5 — Dead Man's Switch	■ Pass ■ Fail ■ Skip	
Test 6 — Position Checklists	■ Pass ■ Fail ■ Skip	

Overall Scenario Result:

■ **PASS** — All tests passed or only minor cosmetic issues noted ■ **CONDITIONAL** — Passes with workaround (document defects above) ■ **FAIL** — Critical defect found (document and halt further testing)

Tester (name / callsign): _____ Date: _____ Time: _____

✓ **Proceed to Scenario 3 only after this scenario passes.**

BETA TEST SCENARIOS

SCENARIO	DURATION	FEATURES TESTED	PAGES EXERCISED
3 of 6	90–120 min	Five ICS sections · T-cards · GPS resource map · All ICS forms · Digital signatures · Planning P · IAP export · Print center · Meetings	ics/* · resources.html · resource_map.html · ics-form.html · iap.html · iap_compile.html · ics213.html · ics214.html · meetings.html · printcenter.html

FOR BETA TESTING USE ONLY — NOT FOR OPERATIONAL USE

This scenario exercises the complete ICS/NIMS incident management workflow from initial briefing through IAP compilation. Every ICS section is visited, every standard form is completed, the T-card resource board is fully populated, GPS positions are set on resources, digital signatures are captured, and the IAP is exported in all three formats: Print, PDF (server-generated), and HTML (fallback).

■ **PREREQUISITE:** An active incident and at least 3 roster members must exist. Use the incident and roster created in Scenarios 1 and 2.

TEST 1 — ICS Overview & Five-Section Navigation (ics/)

Step 1. Navigate to the ICS Overview. Navigate to: http://192.168.50.1/ics/index.html	→ <i>Expected:</i> Five-section navigation bar visible: Overview, Command, Operations, Planning, Logistics, Finance.
Step 2. Click each section in the nav bar: Command, Operations, Planning, Logistics, Finance.	→ <i>Expected:</i> Each section loads correctly. Nav highlights the active section.
Step 3. Navigate to ICS Command. Review the Command Structure section (5.2). Navigate to: http://192.168.50.1/ics/command.html	→ <i>Expected:</i> IC info, incident name, period, objectives visible. Note: select Single IC or Unified Command as appropriate.
Step 4. Navigate to Planning P. Navigate to: http://192.168.50.1/ics/planningp.html	→ <i>Expected:</i> Planning cycle wheel or timeline displays with all 15 phases.

TEST 2 — General Info / ICS-201 Initial Briefing (general_info.html)

Step 5. Navigate to General Info. Navigate to: http://192.168.50.1/general_info.html	→ <i>Expected:</i> ICS-201 form loads with current incident fields.
Step 6. Enter Situation Summary: <i>Beta test exercise — simulated severe weather response. Three shelters activated. No actual emergency.</i>	→ <i>Expected:</i> Text field accepts multi-line input.
Step 7. Enter Initial Actions: <i>Shelter teams deployed. Resource check-in underway.</i>	→ <i>Expected:</i> Field accepts input.
Step 8. Fill in Incident Commander: <i>Beta Tester One</i> and Safety Officer: <i>Beta Tester Two</i> .	→ <i>Expected:</i> Fields accept text.
Step 9. Save the form.	→ <i>Expected:</i> Success confirmation. Data persists on refresh.

TEST 3 — T-Card Resource Board (resources.html)

Step 10. Navigate to the T-Card Resource Board. Navigate to: http://192.168.50.1/resources.html	→ <i>Expected:</i> Board loads. Empty or shows existing resources.
Step 11. Click + Add Resource . Create: Name = Engine 12, Type = Engine, Status = Staging, Assignment = Division Alpha, Leader = KD9TEST, Personnel = 4.	→ <i>Expected:</i> T-card appears on the board in correct status column.
Step 12. Add a second resource: Name = Medical Team 1, Type = Medical Unit, Status = Available, Personnel = 2.	→ <i>Expected:</i> Second T-card appears.
Step 13. Add a third resource: Name = Rescue 3, Type = Crew, Status = Assigned, Assignment = Division Bravo.	→ <i>Expected:</i> Third T-card appears.
Step 14. Click Engine 12's T-card to open the detail panel.	→ <i>Expected:</i> Detail panel opens with INFO and PERSONNEL tabs.
Step 15. Click the PERSONNEL tab.	→ <i>Expected:</i> Personnel tab shows empty list and Add Personnel form.
Step 16. Type <i>Alpha</i> in the Name field.	→ <i>Expected:</i> Roster autocomplete suggests "Alpha Tester".

Step 17. Click the “Alpha Tester” suggestion chip.	→ <i>Expected:</i> Name, callsign, and agency auto-fill from roster.
Step 18. Select ICS position: <i>Crew Boss</i> . Click ✓ Add to Resource .	→ <i>Expected:</i> Alpha Tester added to Engine 12 personnel list. Count badge on T-card updates.
Step 19. Add 2 more personnel to Engine 12.	→ <i>Expected:</i> Personnel count badge shows 3.
Step 20. Remove one person using the X button.	→ <i>Expected:</i> Person removed. Count decrements.
Step 21. Change Engine 12 status to <i>Assigned</i> .	→ <i>Expected:</i> T-card moves to Assigned column on board.
Step 22. Navigate to ICS Operations section. Navigate to: http://192.168.50.1/ics/operations.html	→ <i>Expected:</i> T-card board visible within ICS Operations context.

TEST 4 — GPS-Tracked Resource Map (resource_map.html)

Step 23. Navigate to Resource Map. Navigate to: http://192.168.50.1/resource_map.html	→ <i>Expected:</i> Map loads (dark CartoDB tiles). Select the active incident. Repeater overlay (■■■) available in layer toggle.
Step 24. Verify all three T-cards appear in the sidebar.	→ <i>Expected:</i> Resources with GPS show green left border. Without GPS show dim border.
Step 25. Click Engine 12 in the sidebar to open the Set Position modal.	→ <i>Expected:</i> Modal opens with three placement methods.
Step 26. Click ■ Use Device GPS .	→ <i>Expected:</i> Browser requests location permission. If granted, coordinates fill automatically with accuracy estimate.
Step 27. Enter manual coordinates: Lat = 42.3089, Lon = -88.4356. Add location label: <i>Division Alpha staging, north parking lot</i> .	→ <i>Expected:</i> Coordinate fields accept decimal input.
Step 28. Click Save Position .	→ <i>Expected:</i> Engine 12 pin appears on map. Sidebar entry shows green border and coordinates.
Step 29. Click ■ Click Map to Place for Medical Team 1.	→ <i>Expected:</i> Modal closes. Cursor changes to crosshair.
Step 30. Click anywhere on the map.	→ <i>Expected:</i> Coordinates fill automatically. Modal reopens with the clicked location.
Step 31. Save the position.	→ <i>Expected:</i> Medical Team 1 pin appears on map.
Step 32. Click a resource pin on the map.	→ <i>Expected:</i> Popup shows: resource name, status, assignment, coordinates, GPS source, timestamp, and Update Position button.
Step 33. Click ■ My Location .	→ <i>Expected:</i> Your device location marked on map with amber circle.
Step 34. Enable Auto-refresh 30s .	→ <i>Expected:</i> Map refreshes automatically every 30 seconds.
Step 35. Click Clear GPS on one resource.	→ <i>Expected:</i> Pin removed from map. Sidebar reverts to dim border.

TEST 5 — ICS Form Set with Digital Signatures (ics-form.html)

Complete one entry on each ICS form below. On every form with a Prepared By or Approved By field, capture a digital signature. Save each form and verify data persists.

FORM	KEY FIELDS TO TEST	SIG?	RESULT
ICS-202 Incident Objectives	Enter 3 objectives, tactical priorities, weather	✓ Prepared By	■ Pass ■ Fail
ICS-203 Org Assignment List	Enter IC, OSC, PSC, LSC, FSC names; fill 3 divisions	✓ Prepared By	■ Pass ■ Fail
ICS-204 Assignment List	Add branch, division, assignment, resources needed	✓ Prepared By	■ Pass ■ Fail

FORM	KEY FIELDS TO TEST	SIG?	RESULT
ICS-205 Radio Comms Plan	Add 3 channels from channel library; fill function/frequency	✓ Prepared By	■ Pass ■ Fail
ICS-205A Comms List	Add 3 contact entries		■ Pass ■ Fail
ICS-206 Medical Plan	Add 2 medical facilities; fill trauma level	✓ Prepared By	■ Pass ■ Fail
ICS-207 Org Chart	Verify org chart renders from org assignment data		■ Pass ■ Fail
ICS-208 Safety Message	Enter safety message text; fill hazards	✓ Prepared By	■ Pass ■ Fail
ICS-209 Status Summary	Fill incident status; enter resource summary counts	✓ Prepared By	■ Pass ■ Fail
ICS-211 Check-In List	Verify check-in entries sync from checkin.html		■ Pass ■ Fail
ICS-213 General Message	See Test 6 below	✓ Prepared By	■ Pass ■ Fail
ICS-214 Activity Log	See Test 7 below	✓ Prepared By	■ Pass ■ Fail
ICS-215A Safety Analysis	Enter 3 hazards; fill mitigation measures	✓ Prepared By	■ Pass ■ Fail
ICS-218 Support Vehicle	Add 2 vehicles with mileage/hours		■ Pass ■ Fail
ICS-221 Demob Check-Out	Enter demob instructions; add 1 resource	✓ Prepared By	■ Pass ■ Fail

Digital Signature Test Procedure — apply to every form with SIG REQUIRED:

Step 36. On any ICS form, locate the Prepared By field.	→ <i>Expected:</i> Signature pad canvas appears below the field.
Step 37. Sign using mouse (desktop), finger (touchscreen), or stylus.	→ <i>Expected:</i> Signature appears as drawn stroke in the pad.
Step 38. Click Accept .	→ <i>Expected:</i> Signature preview appears below the field as a PNG image.
Step 39. Save the form. Reload the page.	→ <i>Expected:</i> Signature image persists after reload — restored from database.
Step 40. Click Clear on the signature pad.	→ <i>Expected:</i> Signature removed. Pad is blank.
Step 41. Re-sign and accept.	→ <i>Expected:</i> New signature replaces the cleared one.

TEST 6 — ICS-213 General Message (ics213.html)

Step 42. Navigate to ICS-213 General Message. Navigate to: http://192.168.50.1/ics213.html	→ <i>Expected:</i> Blank message form loads.
Step 43. Fill in: To = Logistics Section Chief, From callsign = KD9TEST (name auto-fills), Subject = Supply Request — Cots, Message = Request 50 cots for Shelter Alpha by 1800.	→ <i>Expected:</i> Callsign auto-fills name from FCC/roster. All fields accept input.
Step 44. Check Reply Requested .	→ <i>Expected:</i> Reply checkbox toggles.
Step 45. Sign Prepared By field.	→ <i>Expected:</i> Signature captured.
Step 46. Click Print .	→ <i>Expected:</i> Print-ready ICS-213 form opens in new tab with all fields filled.
Step 47. Save the message. Verify it appears in the message history.	→ <i>Expected:</i> Message saved and listed.

TEST 7 — ICS-214 Activity Log (ics214.html)

Step 48. Navigate to ICS-214 Activity Log. Navigate to: http://192.168.50.1/ics214.html	→ <i>Expected:</i> Activity log form loads.
Step 49. Select resource: <i>Engine 12</i> .	→ <i>Expected:</i> Resource populates from T-card data.
Step 50. Click + Add Entry . Enter time 0900, Activity = <i>Engine 12 arrived at staging area, crew briefed</i> .	→ <i>Expected:</i> Entry added to log.
Step 51. Add second entry: time 1000, Activity = <i>Deployed to Division Alpha. Crew began structure assessment</i> .	→ <i>Expected:</i> Second entry added.
Step 52. Click Export to FEMA Labor .	→ <i>Expected:</i> Hours from this log push to FEMA Force Account Labor for the active incident.
Step 53. Sign Prepared By field.	→ <i>Expected:</i> Signature captured.
Step 54. Print the ICS-214.	→ <i>Expected:</i> Print-ready log with both entries and signature.

TEST 8 — IAP Assembly & Export (iap.html, iap_compile.html)

FieldCommand provides three IAP output options from iap.html, plus a full PDF compiler at iap_compile.html. Test all three output methods.

Step 55. Navigate to IAP Assembly. Navigate to: http://192.168.50.1/iap.html	→ <i>Expected:</i> IAP summary shows all completed forms for the active incident with completion status indicators.
Step 56. Select operational period and check forms: ICS-202, ICS-203, ICS-204, ICS-205, ICS-206, ICS-208. Click Print IAP . Navigate to: http://192.168.50.1/iap.html	→ <i>Expected:</i> Print-ready IAP opens in new browser tab. Browser print dialog opens automatically.
Step 57. Return to iap.html. Click Save as PDF . Navigate to: http://192.168.50.1/iap.html	→ <i>Expected:</i> Server generates a proper 8.5×11 PDF (via ReportLab). Downloads as IAP_[Incident]_Period[N]_[Date].pdf. Opens correctly in any PDF viewer.
Step 58. Return to iap.html. Click Save as HTML . Navigate to: http://192.168.50.1/iap.html	→ <i>Expected:</i> Browser generates HTML file directly (no server call). File downloads. Open in any browser and use File → Print to print.
Step 59. Navigate to IAP Compile. Select forms: ICS-202, 203, 204, 205, 206, 208. Click Compile IAP PDF . Navigate to: http://192.168.50.1/iap_compile.html	→ <i>Expected:</i> Server generates full PDF (5–15 seconds). Progress indicator shown. PDF contains: title page with incident name and period, section dividers, all six forms in order, embedded signatures.
Step 60. Verify the compiled PDF downloads and opens correctly.	→ <i>Expected:</i> PDF displays all selected forms in FEMA standard order. Signatures embedded in Prepared By fields.

TEST 9 — Meeting Scheduler & Print Center

Step 61. Navigate to Meeting Scheduler. Navigate to: http://192.168.50.1/meetings.html	→ <i>Expected:</i> Meeting calendar/list loads.
Step 62. Create a meeting: Name = Operational Period Briefing, Time = 1800, Location = ICP Main Room.	→ <i>Expected:</i> Meeting saves and appears in list.
Step 63. Navigate to Print Center. Navigate to: http://192.168.50.1/printcenter.html	→ <i>Expected:</i> Document list shows all printable items.
Step 64. Select the IAP PDF and click Print .	→ <i>Expected:</i> Print job routes to configured printer via CUPS, or browser print dialog opens.

SCENARIO 3 — PASS / FAIL SUMMARY

TEST	RESULT	NOTES / DEFECTS
Test 1 — ICS Five-Section Navigation	■ Pass ■ Fail ■ Skip	
Test 2 — General Info / ICS-201	■ Pass ■ Fail ■ Skip	
Test 3 — T-Card Resource Board	■ Pass ■ Fail ■ Skip	
Test 4 — GPS Resource Map	■ Pass ■ Fail ■ Skip	
Test 5 — All ICS Forms & Signatures	■ Pass ■ Fail ■ Skip	
Test 6 — ICS-213 General Message	■ Pass ■ Fail ■ Skip	
Test 7 — ICS-214 Activity Log	■ Pass ■ Fail ■ Skip	
Test 8 — IAP Export (Print / PDF / HTML / Compile)	■ Pass ■ Fail ■ Skip	
Test 9 — Meetings & Print Center	■ Pass ■ Fail ■ Skip	

Overall Scenario Result:

■ PASS — All tests passed or only minor cosmetic issues noted ■ CONDITIONAL — Passes with workaround (document defects above) ■ FAIL — Critical defect found (document and halt further testing)

Tester (name / callsign): _____

Date: _____

Time: _____

✓ Proceed to Scenario 4 only after this scenario passes.

BETA TEST SCENARIOS

SCENARIO	DURATION	FEATURES TESTED	PAGES EXERCISED
4 of 6	60–90 min	Amateur net logger · Public safety net logger · FCC lookup · APRS map · Observer mode · NTS radiogram · Winlink import · AMPRNet · HF propagation · Repeaters · Channel library · ICS-309 · PACTOR modem	netcontrol.html · starcom.html · callsign.html · tactical.html · observer.html · nts.html · winlink-import.html · amprgate.html · propagation.html · repeaters.html · channel_library.html · ics309.html

FOR BETA TESTING USE ONLY — NOT FOR OPERATIONAL USE

This scenario tests all amateur radio emergency communications features: opening and operating both net loggers, FCC callsign validation against the offline database, the tactical APRS map, NTS radiogram generation, Winlink form import, the AMPRNet gateway, HF propagation tool, repeater database, channel library, and ICS-309 export. An FCC amateur license is required for some test steps.

■ **FCC CALLSIGN REQUIRED** for callsign lookup tests. Use your own FCC callsign or any valid active callsign from the FCC ULS database. The offline database contains 800,000+ licensees.

TEST 1 — Amateur Radio Net Control Logger (netcontrol.html)

Step 1. Navigate to Net Control Logger. Navigate to: <code>http://192.168.50.1/netcontrol.html</code>	→ <i>Expected:</i> Net logger loads. Fields: Net Name, Frequency, Mode.
Step 2. Enter: Net Name = <i>Beta Test EMCOMM Net</i> , Frequency = 146.520, Mode = FM.	→ <i>Expected:</i> Fields accept input.
Step 3. Click Open Net .	→ <i>Expected:</i> Net opens with timestamp. Timer starts. Green status banner shows elapsed time.
Step 4. Type your FCC callsign in the Check-In field.	→ <i>Expected:</i> Name auto-fills from offline FCC database. License class shown.
Step 5. Press Enter or click Check In .	→ <i>Expected:</i> Your callsign entry appears in the log with timestamp.
Step 6. Type a second callsign (valid FCC callsign from any licensed amateur).	→ <i>Expected:</i> Name fills automatically.
Step 7. Check in 3 more callsigns. Mix valid and one intentionally invalid callsign.	→ <i>Expected:</i> Valid callsigns fill name. Invalid callsign shows red border or "Not Found".
Step 8. For one check-in, enter remarks: <i>QRM on .520 — switching to .540</i> .	→ <i>Expected:</i> Remarks field accepts text.
Step 9. Click Check Out on one station.	→ <i>Expected:</i> Checkout time recorded. Duration shown rounded to nearest quarter hour.
Step 10. Click ■ Observer Link .	→ <i>Expected:</i> URL copied to clipboard or displayed.
Step 11. Open the observer URL in a second browser tab or different device.	→ <i>Expected:</i> Observer view shows live net status. Updates every 15 seconds. No edit controls visible.
Step 12. Log a traffic entry: Precedence = ROUTINE, From = your callsign, To = BTES EOC, Info = <i>Check-in net closed, 5 stations logged</i> .	→ <i>Expected:</i> Traffic entry appears in the traffic log.
Step 13. Click Close Net .	→ <i>Expected:</i> All remaining stations automatically checked out. Net close time recorded. Duration displayed in header.
Step 14. Click ■ Export ICS-309 .	→ <i>Expected:</i> ICS-309 document downloads. Contains: net name, frequency, mode, open/close times, duration, all check-ins with individual durations (rounded to ¼ hour).

TEST 2 — Public Safety Net Logger (starcom.html)

Step 15. Navigate to Public Safety Net Logger. Navigate to: <code>http://192.168.50.1/starcom.html</code>	→ <i>Expected:</i> Net logger loads. Configured for radio ID check-in.
Step 16. Open a net: Name = <i>Beta Test Starcom Net</i> .	→ <i>Expected:</i> Net opens with timestamp.
Step 17. Check in using radio ID 412 (from ESV-003 in roster).	→ <i>Expected:</i> Name auto-fills from roster by radio ID match.
Step 18. Check in radio ID 999 (not in roster).	→ <i>Expected:</i> Check-in succeeds without roster match. Name field blank for manual entry. Enter name manually.

Step 19. Check in a callsign KD9TEST.	→ <i>Expected:</i> Check-in succeeds. Callsign appears in the log.
Step 20. Close the net and export ICS-309.	→ <i>Expected:</i> ICS-309 downloaded. Radio IDs appear in station column.

TEST 3 — FCC Callsign Lookup (callsign.html)

Step 21. Navigate to Callsign Lookup. Navigate to: <code>http://192.168.50.1/callsign.html</code>	→ <i>Expected:</i> Search field loads.
Step 22. Search for your own callsign.	→ <i>Expected:</i> Returns: callsign, full name, license class, expiration date, grid square, city/state. Instant result — no internet.
Step 23. Search for a callsign that does not exist: ZZ9ZZZ.	→ <i>Expected:</i> "Not found" result displayed.
Step 24. Search for a partial callsign (type 3 letters).	→ <i>Expected:</i> Auto-complete or results filter as you type.
Step 25. Click Add to Roster on a search result.	→ <i>Expected:</i> Roster entry created from FCC data.

TEST 4 — Tactical APRS Map (tactical.html)

Step 26. Navigate to the Tactical Map. Navigate to: <code>http://192.168.50.1/tactical.html</code>	→ <i>Expected:</i> Leaflet map loads with dark tile layer. Controls visible. Repeater overlay (■) available in layer toggle.
Step 27. Verify map tiles load (either online or from local cache).	→ <i>Expected:</i> Map displays geographic features.
Step 28. Enable the Repeater overlay (■). Verify repeater pins appear on map.	→ <i>Expected:</i> Repeater markers visible showing callsign, frequency, and tone on click.
Step 29. If APRS-IS is active (WAN up), verify live stations appear as icons.	→ <i>Expected:</i> APRS station symbols visible on map. Click a station for popup.
Step 30. If RF APRS TNC is connected, verify RF stations appear.	→ <i>Expected:</i> RF stations visible independent of WAN state.
Step 31. Click a station marker to open the popup.	→ <i>Expected:</i> Popup shows: callsign, last beacon time, speed, course, altitude, comment.
Step 32. Navigate to SARTopo Import. Test GeoJSON import with a minimal valid file. Navigate to: <code>http://192.168.50.1/sartopo_import.html</code>	→ <i>Expected:</i> File imports without error. Returns to tactical map.

TEST 5 — NTS Radiogram Generator (nts.html)

Step 33. Navigate to NTS Radiogram. Navigate to: <code>http://192.168.50.1/nts.html</code>	→ <i>Expected:</i> Radiogram form loads with all standard NTS fields.
Step 34. Fill in: Precedence = ROUTINE, Station of origin = your callsign, Place = Testville IL, To = John Smith, Phone = 555-0100.	→ <i>Expected:</i> Fields accept input.
Step 35. Enter text: <i>ALL WELL STOP ARRIVED SAFELY STOP</i>	→ <i>Expected:</i> Word count (check) calculates automatically.
Step 36. Click Generate Radiogram .	→ <i>Expected:</i> Formatted NTS radiogram displays with correct prosign format.
Step 37. Click Print .	→ <i>Expected:</i> Print-ready traffic form opens.

TEST 6 — Winlink Form Import (winlink-import.html)

Step 38. Navigate to Winlink Form Import. Navigate to: <code>http://192.168.50.1/winlink-import.html</code>	→ <i>Expected:</i> Import page loads with file chooser and paste area.
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Step 39. Attempt to import a test file. If no Winlink XML is available, attempt with a plain text file to confirm graceful error handling.

→ *Expected:* Either: import succeeds and ICS form created; OR clear error message if format is invalid. No crash or blank screen.

TEST 7 — AMPRNet (44Net) Gateway (amprgate.html)

■ The AMPRNet gateway requires the 44Net Gateway Pi to be running at 192.168.50.2 and an AMPRNet allocation assigned to your callsign. If gateway Pi is not deployed, skip Steps 40–43 and note as “Gateway not deployed.”

Step 40. Navigate to AMPRNet Gateway Status.

Navigate to: <http://192.168.50.1/amprgate.html>

→ *Expected:* Page loads. Shows tunnel status. Dashboard AMPRNet card shows ■■ Green (tunnel UP) or ■■ Red (tunnel DOWN).

Step 41. Verify public read-only port 9000 is accessible at <http://192.168.50.2:9000>.

→ *Expected:* Callsign login page loads. Enter valid FCC callsign to access status dashboard.

Step 42. Verify the access log shows entries for recent connections.

→ *Expected:* Log shows callsign, IP, timestamp, and result for each access attempt.

Step 43. Attempt to access tunnel control functions from a network device.

→ *Expected:* Control functions require localhost access — port 9001 unavailable from browser. Connection refused.

TEST 8 — HF Propagation, Repeaters & Channel Library

Step 44. Navigate to HF Propagation.

Navigate to: <http://192.168.50.1/propagation.html>

→ *Expected:* Tool loads. If WAN available, current SFI/A/K-index shown with band conditions (10m–80m).

Step 45. If offline, verify last cached data displays.

→ *Expected:* Stale data shown with fetch timestamp. No crash.

Step 46. Navigate to Repeater Database. Import a RepeaterBook CSV (Offline File tab → download from repeaterbook.com free account → drag onto drop zone). Verify map view shows repeater pins (■■ Map View button).

Navigate to: <http://192.168.50.1/repeaters.html>

→ *Expected:* Repeaters load and display. Map view shows pins at correct geographic locations. Click a pin for details.

Step 47. Add a manual test repeater: Callsign = W8RPT, Output = 146.940, Offset = -0.6, CTCSS = 127.3, Mode = FM. Edit it to change CTCSS to 100.0. Delete it.

→ *Expected:* Add, edit, and delete all work correctly.

Step 48. Navigate to Channel Library.

Navigate to: http://192.168.50.1/channel_library.html

→ *Expected:* Pre-loaded interoperability channels visible.

Step 49. Add a custom channel: Name = EOC Primary, Frequency = 155.3400, Mode = FM, Function = Command.

→ *Expected:* Channel saves. Appears in library.

Step 50. Open an ICS-205 form. Verify the channel library picker shows your new channel.

→ *Expected:* Custom channel selectable in ICS-205 channel entry.

SCENARIO 4 — PASS / FAIL SUMMARY

TEST	RESULT	NOTES / DEFECTS
Test 1 — Amateur Net Control Logger	■ Pass ■ Fail ■ Skip	
Test 2 — Public Safety Net Logger	■ Pass ■ Fail ■ Skip	
Test 3 — FCC Callsign Lookup	■ Pass ■ Fail ■ Skip	
Test 4 — Tactical APRS Map	■ Pass ■ Fail ■ Skip	
Test 5 — NTS Radiogram	■ Pass ■ Fail ■ Skip	
Test 6 — Winlink Form Import	■ Pass ■ Fail ■ Skip	

TEST	RESULT	NOTES / DEFECTS
Test 7 — AMPRNet Gateway	■ Pass ■ Fail ■ Skip	
Test 8 — HF Propagation, Repeaters & Channels	■ Pass ■ Fail ■ Skip	

Overall Scenario Result:

■ PASS — All tests passed or only minor cosmetic issues noted ■ CONDITIONAL — Passes with workaround (document defects above) ■ FAIL — Critical defect found (document and halt further testing)

Tester (name / callsign):

Date:

Time:

✓ Proceed to Scenario 5 only after this scenario passes.

BETA TEST SCENARIOS

SCENARIO	DURATION	FEATURES TESTED	PAGES EXERCISED
5 of 6	60–90 min	FEMA labor/equipment/materials · Cost dashboard · NEXRAD radar · Hospitals · Facilities · NIMS resource types · Grid calc · Cheat sheets · Reference library · WAN status	fema_costs.html · fema_rates.html · cost_dashboard.html · radar.html · hospitals.html · facilities.html · resource_types.html · grid.html · cheatsheets.html · refs.html · wan-status.html

FOR BETA TESTING USE ONLY — NOT FOR OPERATIONAL USE

This scenario tests FEMA Public Assistance (PA) cost documentation, the real-time cost dashboard, the animated NEXRAD radar, hospital and facilities directories, NIMS resource typing, the grid square calculator, radio cheat sheets, the offline reference library, and all WAN status tools. Several tests require internet connectivity.

■ NEXRAD radar test requires active internet (WAN) connection. Run offline tests first, then WAN-dependent tests when connected.

TEST 1 — FEMA Equipment Rate Schedule (fema_rates.html)

Step 1. Navigate to FEMA Equipment Rates. Navigate to: http://192.168.50.1/fema_rates.html	→ <i>Expected:</i> Table loads with 44 standard equipment categories and current-year rates.
Step 2. Search for <i>Generator</i> .	→ <i>Expected:</i> Generator entries filter into view.
Step 3. Click — Edit on one entry. Change the rate by \$1.00. Click Save.	→ <i>Expected:</i> Rate updates. Change visible in table.
Step 4. Verify the Rate Year badge shows the current year.	→ <i>Expected:</i> Badge shows current year. No stale-data warning.

TEST 2 — FEMA PA Cost Documentation — Labor (fema_costs.html)

Step 5. Navigate to FEMA PA Cost Documentation. Navigate to: http://192.168.50.1/fema_costs.html	→ <i>Expected:</i> Three tabs visible: Force Account Labor, Equipment, Materials/Contracts.
Step 6. Click + Add Labor Entry. Enter: Employee = Alpha Tester, Title = Emergency Management Coordinator, Regular Hours = 8, Overtime Hours = 4, Hourly Rate = 32.50, Fringe % = 28.	→ <i>Expected:</i> Fields accept numeric and text input.
Step 7. Verify the Total with Fringe calculates automatically: $(8+4) \times \$32.50 \times 1.28 = \499.20	→ <i>Expected:</i> Calculated total matches expected value.
Step 8. Save the labor entry.	→ <i>Expected:</i> Entry saved. Appears in labor list.
Step 9. Add a second labor entry with different fringe rate.	→ <i>Expected:</i> Both entries appear. Individual totals correct.
Step 10. Click Import from ICS-214.	→ <i>Expected:</i> ICS-214 import dialog or picker opens.

TEST 3 — FEMA PA Cost Documentation — Equipment

Step 11. Click the Equipment tab.	→ <i>Expected:</i> Equipment cost form loads.
Step 12. Click + Add Equipment Entry. Click ■ Lookup.	→ <i>Expected:</i> FEMA rate picker opens. Searchable list of 44 equipment categories.
Step 13. Search for <i>Generator</i> . Select a 15 kW generator entry.	→ <i>Expected:</i> FEMA rate fills automatically from the rate schedule.
Step 14. Enter: Hours = 16, Equipment ID = GEN-001.	→ <i>Expected:</i> Total calculates: $16 \times (\text{FEMA rate})$.
Step 15. Save the equipment entry.	→ <i>Expected:</i> Entry saved. Total shown.

TEST 4 — FEMA PA Cost Documentation — Materials & Contracts

Step 16. Click the Materials tab. Add: Vendor = Home Depot, Description = Sandbags 50lb, PO = PO-2026-001, Unit Cost = 4.50, Qty = 200.	→ <i>Expected:</i> Total calculates: $200 \times \$4.50 = \900.00
Step 17. Save materials entry.	→ <i>Expected:</i> Entry saved.
Step 18. Click the Contracts tab. Add: Vendor = Cleanup Contractors LLC, Description = Debris removal 4 hours, Contract # = C-001, Amount = 2400.00.	→ <i>Expected:</i> Entry saved with contract number.

Step 19. Click ■ Export PW Text .	→ <i>Expected:</i> Project Worksheet text downloads. Contains itemized costs by category and total claimed amount.
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TEST 5 — Real-Time Cost Dashboard (cost_dashboard.html)

Step 20. Navigate to Cost Dashboard. Navigate to: <code>http://192.168.50.1/cost_dashboard.html</code>	→ <i>Expected:</i> All cost panels load showing data from Tests 2–4.
Step 21. Verify Total Incident Cost tile shows the sum of all entries.	→ <i>Expected:</i> Total = Labor total + Equipment total + Materials total + Contracts total.
Step 22. Verify Cost by Category breakdown bars are proportional.	→ <i>Expected:</i> Labor, Equipment, Materials, Contracts bars visible with \$ amounts and %.
Step 23. Enter a budget amount in the Budget Tracker panel.	→ <i>Expected:</i> Remaining balance and percentage used calculate correctly.
Step 24. Verify the Per-Period Breakdown shows costs attributed to the active period.	→ <i>Expected:</i> Period 1 costs listed.
Step 25. Wait 2 minutes. Verify dashboard auto-refreshes.	→ <i>Expected:</i> Data refreshes without manual reload.

TEST 6 — Animated NEXRAD Radar (radar.html) — WAN REQUIRED

■ This test requires active internet. If offline, verify the offline banner appears and test the fallback display only.

Step 26. Navigate to NEXRAD Radar. Navigate to: <code>http://192.168.50.1/radar.html</code>	→ <i>Expected:</i> Page loads. If WAN up: radar tiles begin loading. If WAN down: offline banner shown.
Step 27. Select your nearest NEXRAD station from the dropdown.	→ <i>Expected:</i> Radar station changes. New tiles load for selected station.
Step 28. Click ■ Play .	→ <i>Expected:</i> Radar animates as a loop.
Step 29. Click ■ Pause .	→ <i>Expected:</i> Animation stops on current frame.
Step 30. Use the timeline scrubber to jump to an earlier frame.	→ <i>Expected:</i> Radar updates to that frame.
Step 31. Change speed to 2×.	→ <i>Expected:</i> Animation plays faster.
Step 32. Switch palette to Dual-pol.	→ <i>Expected:</i> Color scheme changes.
Step 33. Click ■ Step backward one frame.	→ <i>Expected:</i> Moves exactly one frame back.
Step 34. Verify the reflectivity legend is visible.	→ <i>Expected:</i> Color scale with dBZ values shown.
Step 35. Turn off WAN (unplug ethernet/disable hotspot). Reload the radar page.	→ <i>Expected:</i> Offline banner appears. Last loaded frame displayed with timestamp.

TEST 7 — Hospital Proximity & Facilities Directory

Step 36. Navigate to Hospital Proximity. Navigate to: <code>http://192.168.50.1/hospitals.html</code>	→ <i>Expected:</i> Hospital directory loads.
Step 37. Add a hospital: Name = Beta General Hospital, Address = 100 Main St Testville IL, Trauma Level = Level II, Helipad = Yes.	→ <i>Expected:</i> Hospital saves and appears in list.
Step 38. Navigate to Facilities Directory. Navigate to: <code>http://192.168.50.1/facilities.html</code>	→ <i>Expected:</i> Facilities list loads.
Step 39. Add a facility: Name = Incident Command Post, Type = ICP, Address = 200 Park Ave Testville IL, Capacity = 50 personnel.	→ <i>Expected:</i> Facility saves. Appears in list.

Step 40. Edit the facility. Add Notes: *Generator on site. Porta-johns deployed.*

→ *Expected:* Notes save correctly.

TEST 8 — NIMS Resource Typing Library (resource_types.html)

Step 41. Navigate to NIMS Resource Types.
Navigate to: http://192.168.50.1/resource_types.html

→ *Expected:* Pre-loaded NIMS types visible: Engine, Crew, Helicopter, etc.

Step 42. Add a custom resource type: Name = Amateur Radio Go-Box, Category = Communications.

→ *Expected:* Custom type saves. Appears in list.

Step 43. Navigate to T-Card board. Click + **Add Resource**. Verify new type appears in the Type dropdown.
Navigate to: <http://192.168.50.1/resources.html>

→ *Expected:* Custom type “Amateur Radio Go-Box” selectable.

Step 44. Delete the custom resource type from the type library.
Navigate to: http://192.168.50.1/resource_types.html

→ *Expected:* Type removed from library.

TEST 9 — Grid Calculator, Cheat Sheets, Reference Library & WAN Status

Step 45. Navigate to Grid Square Calculator.
Navigate to: <http://192.168.50.1/grid.html>

→ *Expected:* Calculator loads.

Step 46. Enter grid square EN52ab. Click Convert.

→ *Expected:* Decimal coordinates returned: approx 42.31°N, 88.44°W.

Step 47. Enter two grid squares and calculate distance/bearing.

→ *Expected:* Distance in km and miles plus true bearing returned.

Step 48. Navigate to Radio/ICS Cheat Sheets.
Navigate to: <http://192.168.50.1/cheatsheets.html>

→ *Expected:* Quick reference cards visible: phonetic alphabet, Q-codes, RST scale, ICS structure, NIMS resource types, HF calling frequencies.

Step 49. Click Print on any cheat sheet card.

→ *Expected:* Print layout opens. Card fits on one page.

Step 50. Navigate to Reference Library (Kiwix).
Navigate to: <http://192.168.50.1/refs.html>

→ *Expected:* Kiwix interface loads. Offline content available.

Step 51. Navigate to WAN Status detail page.
Navigate to: <http://192.168.50.1/wan-status.html>

→ *Expected:* Detailed WAN status shows both configured sources, detection method, role, last check time, and current state.

Step 52. Verify both WAN sources are listed with their roles.

→ *Expected:* Preferred and Fallback sources both displayed. Your group's chosen primary and secondary clearly identified.

SCENARIO 5 — PASS / FAIL SUMMARY

TEST	RESULT	NOTES / DEFECTS
Test 1 — FEMA Equipment Rates	■ Pass ■ Fail ■ Skip	
Test 2 — FEMA Force Account Labor	■ Pass ■ Fail ■ Skip	
Test 3 — FEMA Equipment Cost	■ Pass ■ Fail ■ Skip	
Test 4 — FEMA Materials & Contracts	■ Pass ■ Fail ■ Skip	
Test 5 — Cost Dashboard	■ Pass ■ Fail ■ Skip	
Test 6 — NEXRAD Radar	■ Pass ■ Fail ■ Skip	
Test 7 — Hospitals & Facilities	■ Pass ■ Fail ■ Skip	
Test 8 — NIMS Resource Types	■ Pass ■ Fail ■ Skip	

TEST	RESULT	NOTES / DEFECTS
Test 9 — Grid, Cheat Sheets, Refs, WAN	☐ Pass ☐ Fail ☐ Skip	

Overall Scenario Result:

☐ PASS — All tests passed or only minor cosmetic issues noted

☐ CONDITIONAL — Passes with workaround (document defects above)

☐ FAIL — Critical defect found (document and halt further testing)

Tester (name / callsign): _____

Date: _____

Time: _____

✓ Proceed to Scenario 6 only after this scenario passes.

BETA TEST SCENARIOS

SCENARIO	DURATION	FEATURES TESTED	PAGES EXERCISED
6 of 6	90–120 min	Multi-device access · WAN failover under load · Sustained 60-min simulation · Edge cases · Error handling · Performance measurements · Final cleanup	3 devices minimum · 2 operators minimum · All major tools

FOR BETA TESTING USE ONLY — NOT FOR OPERATIONAL USE

This is the final and most demanding scenario. It simulates a real multi-operator activation with several devices simultaneously accessing the system, exercises WAN failover under load, tests every edge case and error condition, and verifies system stability over a sustained 2-hour operational period. Minimum 2 operators and 3 devices required.

■ **PREREQUISITE:** All previous scenarios must pass before running Scenario 6. This scenario assumes a fully configured system with roster, channels, repeaters, and at least one completed incident from prior scenarios.

■ **SETUP:** Assign roles before starting. Suggested: Device 1 = Incident Commander / Net Control; Device 2 = Operations / T-Cards / Resource Map; Device 3 = Logistics / FEMA Costs / Check-In Station. All three devices connect to EMCOMM-NET simultaneously.

TEST 1 — Multi-Device Simultaneous Access

Step 1. Connect all 3 devices to EMCOMM-NET simultaneously.	→ <i>Expected:</i> All devices receive IP addresses via DHCP. Network shows 3+ connected clients.
Step 2. On each device, navigate to <code>http://192.168.50.1</code> simultaneously.	→ <i>Expected:</i> All three dashboards load without delay. No timeout or error.
Step 3. On Device 1: open the Net Control Logger and start a net. Navigate to: <code>http://192.168.50.1/netcontrol.html</code>	→ <i>Expected:</i> Net opens on Device 1.
Step 4. On Device 2: open the T-Card Resource Board. Navigate to: <code>http://192.168.50.1/resources.html</code>	→ <i>Expected:</i> T-Card board loads on Device 2.
Step 5. On Device 3: open Personnel Accountability. Navigate to: <code>http://192.168.50.1/accountability.html</code>	→ <i>Expected:</i> Accountability page loads on Device 3.
Step 6. While all three tools are open and active, perform simultaneously: Device 1 logs a check-in; Device 2 changes a T-card status; Device 3 refreshes the PAR page.	→ <i>Expected:</i> All three operations complete without conflict. No 500 errors, no database lock errors, no timeouts.
Step 7. On Device 2, open the Observer Link from Device 1's net.	→ <i>Expected:</i> Observer view loads on Device 2 and shows live net status.

TEST 2 — WAN Failover Under Load

Step 8. Verify WAN is active (green status on dashboard).	→ <i>Expected:</i> Primary WAN indicator green.
Step 9. Load the NEXRAD Radar page on Device 3 and start playback. Navigate to: <code>http://192.168.50.1/radar.html</code>	→ <i>Expected:</i> Radar animates normally.
Step 10. Disconnect the primary WAN source (unplug antenna or disable hotspot).	→ <i>Expected:</i> Dashboard WAN status changes to Fallback (blue) or Offline (red) within 60 seconds. Radar page shows offline banner.
Step 11. Verify all ICS tools on Devices 1 and 2 continue working.	→ <i>Expected:</i> Net logger, T-cards, check-in, accountability — all fully functional offline.
Step 12. Reconnect primary WAN.	→ <i>Expected:</i> Dashboard WAN status returns to primary (green) within 60 seconds. Radar resumes automatically.
Step 13. Repeat failover 3 times in succession.	→ <i>Expected:</i> System recovers correctly each time. No data loss. No session drops.

TEST 3 — Sustained 60-Minute Operational Period Simulation

Run the following activities continuously for 60 minutes as a simulated real activation. Record any errors, slowdowns, or unexpected behavior.

MINUTE	ACTIVITY	DEVICE	RESULT
0–5	Net Control: Open net and log 5 check-ins	1	■ OK ■ Issue

MINUTE	ACTIVITY	DEVICE	RESULT
5–10	Operations: Add 3 resources to T-card board	2	■ OK ■ Issue
10–15	Check-In: QR scan check-in for 3 personnel	3	■ OK ■ Issue
15–20	IC: Complete ICS-202 objectives	1	■ OK ■ Issue
20–25	Operations: Set GPS position on 2 resources	2	■ OK ■ Issue
25–30	Conduct PAR on all checked-in personnel	3	■ OK ■ Issue
30–35	Net Control: Log traffic entry; export ICS-309	1	■ OK ■ Issue
35–40	Finance: Add 2 FEMA labor entries	2	■ OK ■ Issue
40–45	IC: Compile IAP PDF (ICS-202, 203, 205)	1	■ OK ■ Issue
45–50	Operations: Update T-card status changes	2	■ OK ■ Issue
50–55	Net Control: Close net; open new period	1	■ OK ■ Issue
55–60	All devices: Navigate dashboard; verify no lag	All	■ OK ■ Issue
Total errors/issues during sustained period: _____			■ Fast ■ OK ■ Slow ■ Unacceptable

TEST 4 — Edge Cases & Error Handling

Test each edge case. Record whether the system handles it gracefully (no crash, clear error message, data preserved).

EDGE CASE	HOW TO TEST	EXPECTED BEHAVIOR	RESULT
Empty incident — no data	Create new incident, immediately open IAP Compile	Empty state shown gracefully. No crash.	■ Pass ■ Fail
Very long text in form fields	Enter 500+ character string in ICS-202 objectives	Text accepted. Truncated or wrapped in PDF. No crash.	■ Pass ■ Fail
Special characters in fields	Enter: & " ' < > / \ # @ ! in roster name	Characters stored and displayed correctly. No XSS or SQL error.	■ Pass ■ Fail
Duplicate callsign check-in	Check in the same callsign twice to the same net	Either: duplicate warning shown, OR both entries logged with timestamps.	■ Pass ■ Fail
Check out before checking in	Try to click Check Out on a person already checked out	Button disabled or graceful no-op. No duplicate checkout.	■ Pass ■ Fail
Invalid coordinates in GPS	Enter lat = 999, lon = 999 in resource map modal	Validation error: "Coordinates out of range." No pin placed.	■ Pass ■ Fail
Zero-cost FEMA entries	Add labor entry with 0 hours and 0 rate	Entry saves. Total = \$0.00. No division-by-zero error.	■ Pass ■ Fail
Delete active incident	Attempt to delete/archive the currently active incident	Either: warning shown before proceeding, OR deleted cleanly.	■ Pass ■ Fail
QR scan — camera permission denied	Click Start Camera then deny permission	Clear error message. Manual entry fallback still usable.	■ Pass ■ Fail
Very large roster import (50+ members)	Import a CSV with 50 entries	All 50 import successfully. No timeout. Page remains responsive.	■ Pass ■ Fail
Simultaneous form saves from two devices	Two devices edit and save the same ICS-202 form at the same time	Last-write wins. No crash. Data coherent.	■ Pass ■ Fail
API server unreachable (localStorage fallback)	Kill the API server from the Pi terminal. Try to use a form.	localStorage fallback activates. Warning shown. Data preserved locally.	■ Pass ■ Fail
Disconnect from EMCOMM-NET mid-session	Disconnect Wi-Fi on one device. Reconnect. Resume work.	Page reloads and recovers. No data lost.	■ Pass ■ Fail
Browser back button from ICS form	Fill out ICS-205, navigate back, navigate forward	Data retained. No data loss from browser navigation.	■ Pass ■ Fail

TEST 5 — ICS-204A Briefing Sheet & Starcom Resource Map

Step 14. Navigate to ICS-204A Briefing Sheet. Navigate to: http://192.168.50.1/briefing_204a.html	→ <i>Expected:</i> Form loads with assignment briefing fields.
Step 15. Fill in Branch, Division, operational period, and 2 resources assigned.	→ <i>Expected:</i> Fields accept input.
Step 16. Save and print.	→ <i>Expected:</i> Print layout opens with all data.
Step 17. Navigate to Starcom Resource Map. Navigate to: http://192.168.50.1/resmap.html	→ <i>Expected:</i> Public safety resource map loads. Different from ICS resource map.
Step 18. Add a Starcom unit position to the map.	→ <i>Expected:</i> Unit marker appears on map.

TEST 6 — System Performance Measurements

Measure and record the following performance metrics. These establish a baseline for v1.0 and identify any performance issues.

MEASUREMENT	HOW TO MEASURE	RESULT (sec)	ACCEPTABLE?
Dashboard load time	Hard refresh (Ctrl+Shift+R) on index.html. Time to full load.	____s	■ Yes (<3s) ■ No
FCC callsign lookup	Time from typing callsign to name appearing in Net Logger.	____s	■ Yes (<1s) ■ No
T-card board with 10 resources	Time for T-card board to load with 10 resources.	____s	■ Yes (<2s) ■ No
IAP PDF compilation (6 forms)	Time from clicking Compile to PDF download beginning.	____s	■ Yes (<20s) ■ No
IAP Save as PDF (iap.html)	Time from clicking Save as PDF to PDF download beginning.	____s	■ Yes (<15s) ■ No
Personnel Accountability load	Time for PAR page to load with 10 check-ins.	____s	■ Yes (<3s) ■ No
QR scan detection time	Time from holding QR to camera to detection confirmation.	____s	■ Yes (<2s) ■ No
GPS resource map with 5 pins	Time for map to load and display 5 resource markers.	____s	■ Yes (<3s) ■ No
Cost dashboard with 10 entries	Time for cost dashboard to load and calculate totals.	____s	■ Yes (<3s) ■ No

TEST 7 — Final Cleanup & Beta Reset

Step 19. Archive all test incidents to USB before reset. Navigate to: http://192.168.50.1/incident_mgmt.html	→ <i>Expected:</i> JSON archive files written to USB for records.
Step 20. Run Beta/Scenario Reset.	→ <i>Expected:</i> All test data cleared. Roster and config preserved.
Step 21. Verify the system is in clean state for handover or next activation.	→ <i>Expected:</i> Dashboard shows no active incident. Roster intact.
Step 22. Document any unresolved defects in the Defect Log on the final page.	→ <i>Expected:</i> All defects captured for developer review.

SCENARIO 6 — PASS / FAIL SUMMARY

TEST	RESULT	NOTES / DEFECTS
Test 1 — Multi-Device Simultaneous Access	■ Pass ■ Fail ■ Skip	
Test 2 — WAN Failover Under Load	■ Pass ■ Fail ■ Skip	
Test 3 — Sustained 60-Minute Simulation	■ Pass ■ Fail ■ Skip	
Test 4 — Edge Cases & Error Handling	■ Pass ■ Fail ■ Skip	
Test 5 — 204A Briefing & Starcom Map	■ Pass ■ Fail ■ Skip	
Test 6 — Performance Measurements	■ Pass ■ Fail ■ Skip	
Test 7 — Final Cleanup	■ Pass ■ Fail ■ Skip	

Overall Scenario Result:

■ PASS — All tests passed or only minor cosmetic issues noted ■ CONDITIONAL — Passes with workaround (document defects above) ■ FAIL — Critical defect found (document and halt further testing)

Tester (name / callsign): _____

Date: _____

Time: _____

OVERALL BETA TEST SUMMARY — ALL SIX SCENARIOS

SCENARIO	TITLE	RESULT	CRITICAL DEFECTS
1	System Setup & Incident Creation	■ Pass ■ Fail	
2	Roster, Check-In & Personnel Accountability	■ Pass ■ Fail	
3	Full ICS Platform, T-Cards & IAP	■ Pass ■ Fail	
4	Amateur Radio EMCOMM & Net Loggers	■ Pass ■ Fail	
5	FEMA Documentation & Situational Awareness	■ Pass ■ Fail	
6	Multi-Operator Simulation & Stress Test	■ Pass ■ Fail	

FINAL BETA TEST VERDICT:

■ PASS — System ready for operational deployment ■ CONDITIONAL PASS — Passes with documented workarounds. Critical defects must be resolved before operational use. ■ FAIL — Critical defects prevent operational deployment.

DEFECT LOG

Document all defects found during testing. Include: Scenario #, Test #, Step #, description of behavior, expected vs. actual, and severity.

ID	SCENARIO / TEST / STEP	DESCRIPTION	SEVERITY	STATUS
D-001			■ Critical ■ Major ■ Minor	■ Open ■ Fixed
D-002			■ Critical ■ Major ■ Minor	■ Open ■ Fixed
D-003			■ Critical ■ Major ■ Minor	■ Open ■ Fixed
D-004			■ Critical ■ Major ■ Minor	■ Open ■ Fixed
D-005			■ Critical ■ Major ■ Minor	■ Open ■ Fixed
D-006			■ Critical ■ Major ■ Minor	■ Open ■ Fixed
D-007			■ Critical ■ Major ■ Minor	■ Open ■ Fixed
D-008			■ Critical ■ Major ■ Minor	■ Open ■ Fixed

Lead Beta Tester:		Callsign:	
Organization:		Date:	

Submit completed beta test package (photo or scan) to: **James Rospopo KE4CON** ·
<https://github.com/KE4CON/FieldCommand-IMS/issues>

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